School of Information Technology
AURO University, Surat, Gujarat.

Curriculum
for
BSC-IT SEM 3 and
Integrated MSC
IT SEM – 3
(Batch- 2022-25)

School of Information Technology
AURO University
Program Outcomes (PO)

Learners are expected to know and be able to–

PO 01	Knowledge	Ability to build (either independently or by joining a higher academic program) one of the core IT concepts learned in the course. Ability to apply the core computer science concepts to solve problems in the I'I' industry
PO 02	Problem Analysis	Students are equipped with skills to solve computational problems at their workplace and for the society
PO 03	Professional Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the IT practice.
PO 04	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams. and in multidisciplinary settings
PO 05	Modern Tool Usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling to complex Engineering activities, with an understanding of the limitations.
PO 06	Communication	Students demonstrate effective communication presentation skills while interacting with professional peers and in society
PO 07	Scientific Reasoning	Given a problem. the students will be able to analyze it, suggest solutions, and critically evaluate the solutions proposed by others
PO 08	Integral Learning	Recognize the need for, and have the preparation and ability to engage in independent and lifelong integrated learning in the broadest context of technological change

AURO University BSC IT and IMSC IT SEM 3 (Batch 2022-25) IJAPT0301 - Java Programming		
Teaching Scheme:	**Credit**	**Evaluation Scheme:**
04 Hours/Week	**04**	**Internal Assessment(TH): 50 Marks** **End Semester(TH): 50 Marks**

Prerequisite Courses, if any: Basic Knowledge of Programming

Companion Course, if any: ---

Course Objectives:

- To understand the basic concept of Java programming
- To learn how to implement object-oriented programming concepts of Java like Inheritance, Encapsulation, polymorphism, and abstraction.
- To learn how to make an application with threads in Java and also how to manage it.
- To learn how to develop GUI-based Applications and Connect with Databases using Java.

Course Outcomes:

On completion of the course, the learner will be able to–

CO1: Will be able to understand the fundamental concept of Java and object-oriented programming with its features.

CO2: Will be able to understand and Demonstrate the concept of the class, package, and manage runtime errors.

CO3: Will be able to Demonstrate Multi-tasking using multiple threads and manage exception handling to develop efficient and error-free code

CO4: Will be able to understand and demonstrate graphical user interface (GUI) Applications using JDBC.

Course Contents

Unit I	**Introduction of Java and OOPs Programming Concepts**	**(16 Hours)**

What is Java? Why Java, History of Java, Buzzwords of Java, Principals of OOPS, JDK, JRE, JVM, Setup Java Environment, First Java Program, Variables, Keywords, Data Types, Type Casting of Data Type, Arrays, Operators, Decision Control Structure, Loop Control Structure, Class, Object and Method, OOPS (Polymorphism, Inheritance, Encapsulation), Abstract Class.

Mapping of Course Outcomes for Unit I	CO1	Blooms Level: 1,2

Unit II	**Class, Package, Interface, Exception Handling**	**(16 Hours)**

What is Class, Constructor, Garbage Collection, Keyword – this, final, finalize, Inheritance, abstract class, What is Package, Create Package, Import Package, Define Interface, Nested Interface, what is Exception, Expectation Type, Try and Catch Block, Nested Try Block, throw, throws, finally, User Define Exception

Mapping of Course Outcomes for Unit II	CO2	Blooms Level: 2,3

Unit III	Multithreading Programming and String and I/O Handling Operation	(16 Hours)

Introduction of thread and runnable interface, create a single thread and multithread application, use of is Alive and Join Method, inter-thread Communication, Wrapper Class, Autoboxing and auto unboxing, String Handling in Java, The Collection Framework – The Collection Interfaces, The Collection classes. Utility Classes - String Tokenizer, I/O Programming in Java

Mapping of Course Outcomes for Unit III	CO3	Blooms Level: 2,3,4

Unit IV	GUI Programming, JDBC, and CRUD Operation	(16 Hours)

GUI Programming- AWT, Component of AWT, Type of Listener and Event, Basics of JDBC, Build First Desktop application using JDBC with all CRUD operation

Mapping of Course Outcomes for Unit IV	CO4	Blooms Level: 3,5,6

*1=knowledge; 2=Comprehension/ Understanding; 3 =Application; 4=Analysis; 5=Evaluation; 6=Creating

Learning Resources

Text Books:

1. Java – The Complete Reference, Herbert Schildt, McGraw-Hill Education

Reference Books:

1. E Balagurusamy, "Programming with JAVA", McGraw-Hill Education
2. Core and advanced Java Black book, Dreamtech
3. Head First Java, Kathy Sierra, O'Reilly publication

e-Books

MOOC Courses:

@The CO-PO Mapping Table

CO's/ PO's	PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08
CO 01	3	-	1	-	2	3	-	2
CO 02	-	2	2	-	-	2	1	1
CO 03	-	3	2	-	2	-	-	-
CO 04	2	3	-	2	3	2	3	2

*3 – HIGH, 2 – Medium, 1 - LOW

Sr No	Assessment Tools	Total Marks	Weightage	Remarks
1	Assignments	15	15%	Assignments will be shared and the same will be discussed in class submission starting from the 6th week onwards.
2	Mid Term	20	20%	Mid Term Examination related to the topics covered in the previous weeks (10th week)
3	Lab Test	15	15%	Students need to give a Practical Lab Test based previous week's learning (15th Week)
4	End – Term Examination	50	100%	

Assessment and Evaluation

Rubric for an Assignment that assigns a total of 15 marks:

Criterion	Description	Marks
Criterion 1: Content	Relevancy information to the question and all aspects of the question fully addressed.	8
Criterion 2: Organization	Skilfully organized Content with examples and figures.	3
Criterion 3: Delivery	Clarity and Confidence: The presenter speaks clearly, confidently, and maintains eye contact with the audience, and effective use of hand and body gesture	2
Criterion 4: Grammar & Mechanics	An essay is free of distracting spelling, punctuation, and grammatical errors.	2

Rubric for a Lab Test that assigns a total of 15 marks:

Criterion	Description	Marks
Criterion 1: Accuracy and Completion of task	Programs are written precisely based on the question with accuracy	7
Criterion 2: Complexity	The Code is well-organized and very easy to follow.	3
Criterion 3: Result or Output	Represent all data appropriately. The findings of the results are clear.	3
Criterion 3: Completion of the target in Lab	target has been completed within a time limit	2